Safe, Reliable, and Robust Autonomous Operations at NASA Langley Research Center's Autonomy Incubator

B. Danette Allen, PhD
Head, Autonomy Incubator
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Presented by: Carlos G. Dávila

Rising to the Autonomy Challenge



http://www.nasa.gov/larc/nasa-langley-autonomy-incubator-leading-the-push-for-intelligent-machines/

Intelligence and Autonomy

The 'A' Word



http://aviationweek.com/commercial-aviation/certifiable-trust-required-take-autonomous-systems-past-unmanned

Automation vs. Autonomy

"There is a paradigm shift from automated to autonomous: **automation is relegation; autonomy is delegation**…"

"… safe and trusted systems than can perceive their environment for situational awareness and assessment, make decisions on uncertain and inaccurate information, act appropriately, learn from experience and adapt their behavior…"

"…[certification] is about behavior and probability… we will need new methods of verification and validation."

Three Goals

1. Build a Multi-Disciplinary Team

- Mechanics
- Electronics
- Controls
- Programming
- Psychology
- Signal Processing
- Computer Science
- Computer Vision
- Machine Learning
- …

2. Enable new missions in

- Space
- Aero
- Science

3. Create a Testbed for Autonomous Systems

- **Open Software Architecture (AEON)**
- **CERTAIN**

Autonomy Challenges

- **Human-Machine Interaction/Teaming**
- **Data-rich/degraded/deprived environments**
- Size, Weight And Power (SWAP)
- Sensor Fusion
- Adaptive Control
- Geo-containment
- **Sense/Detect and Avoid (DAA)**
- Object Detection and Classification
- Separation Management
- **Precision navigation**
- **Localization**
- Landing site selection
- **Adaptation and Learning**
- Performance Standards
- Verification and Validation (V&V)
- Certification
- Trust
- Communication, C2
- **Test and Evaluation (T&E)**

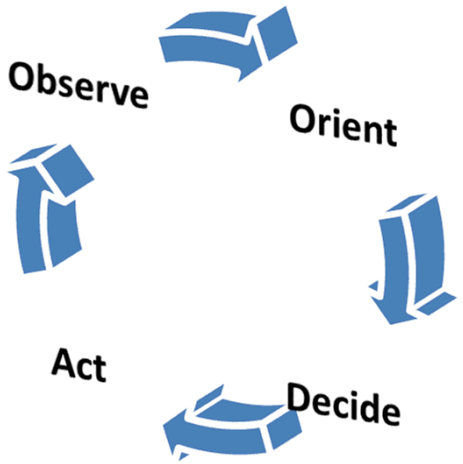
Bridging the Chasm:

NASA's Missions and Autonomy

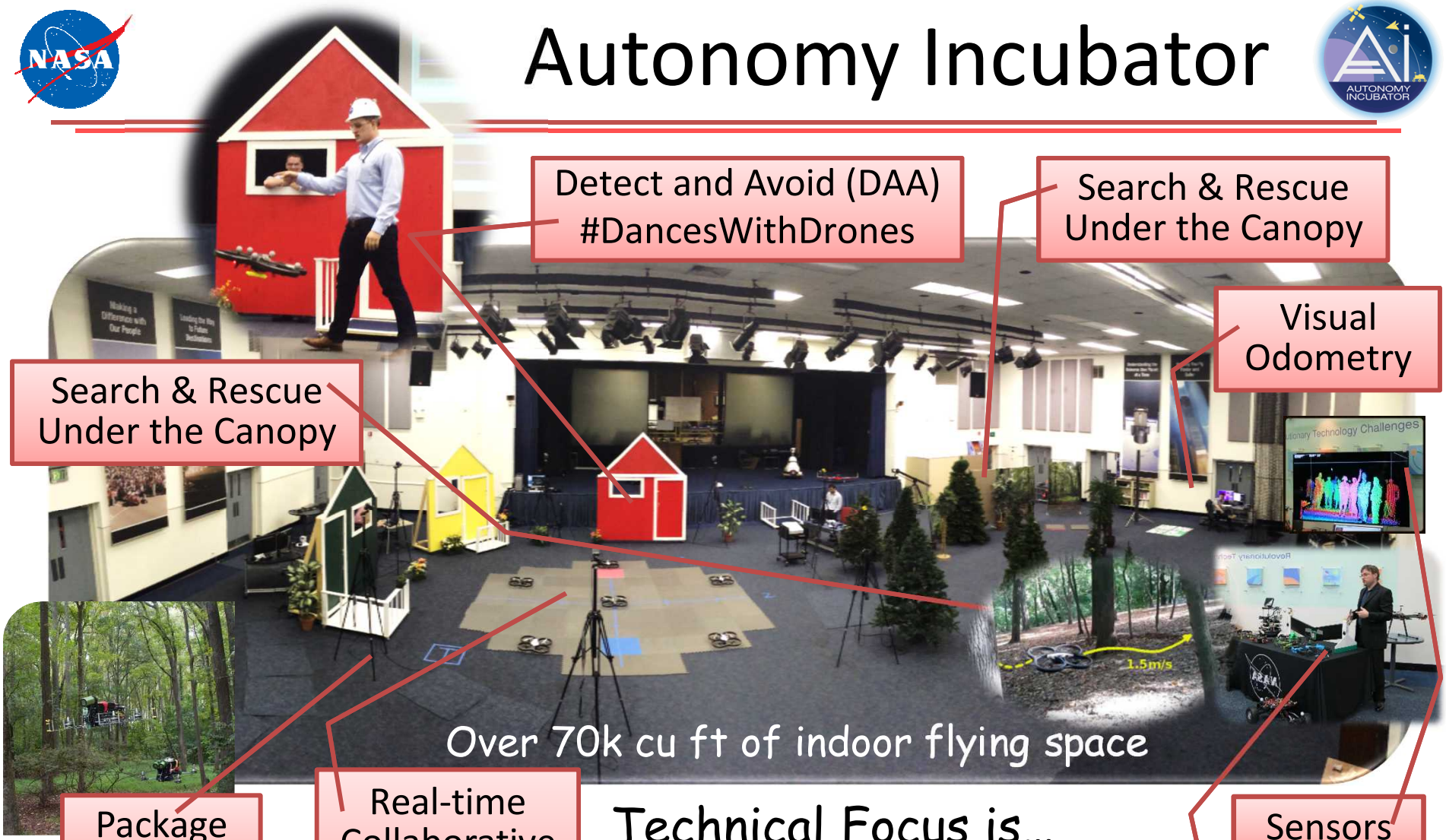
- As we move towards On-Demand Mobility (ODM) in aeronautics and beyond LEO and L2 in space exploration, **human intelligence** applied to supervision, control, and intervention of operations will **no longer** be **viable** due to system/mission complexity, short reaction/decision time, comm delays, distance, hostile environments, and close proximity.

- This requires that we design, build, and test systems capable of responding to expected and unexpected situations with **machine intelligence** similar to that of humans. This means
  - trusted and certified-safe systems capable of
  - sensing and perception
  - situation assessment/awareness
  - decision-making
  - taking action
  - and knowledge acquisition (learning)

  that **team with humans**.

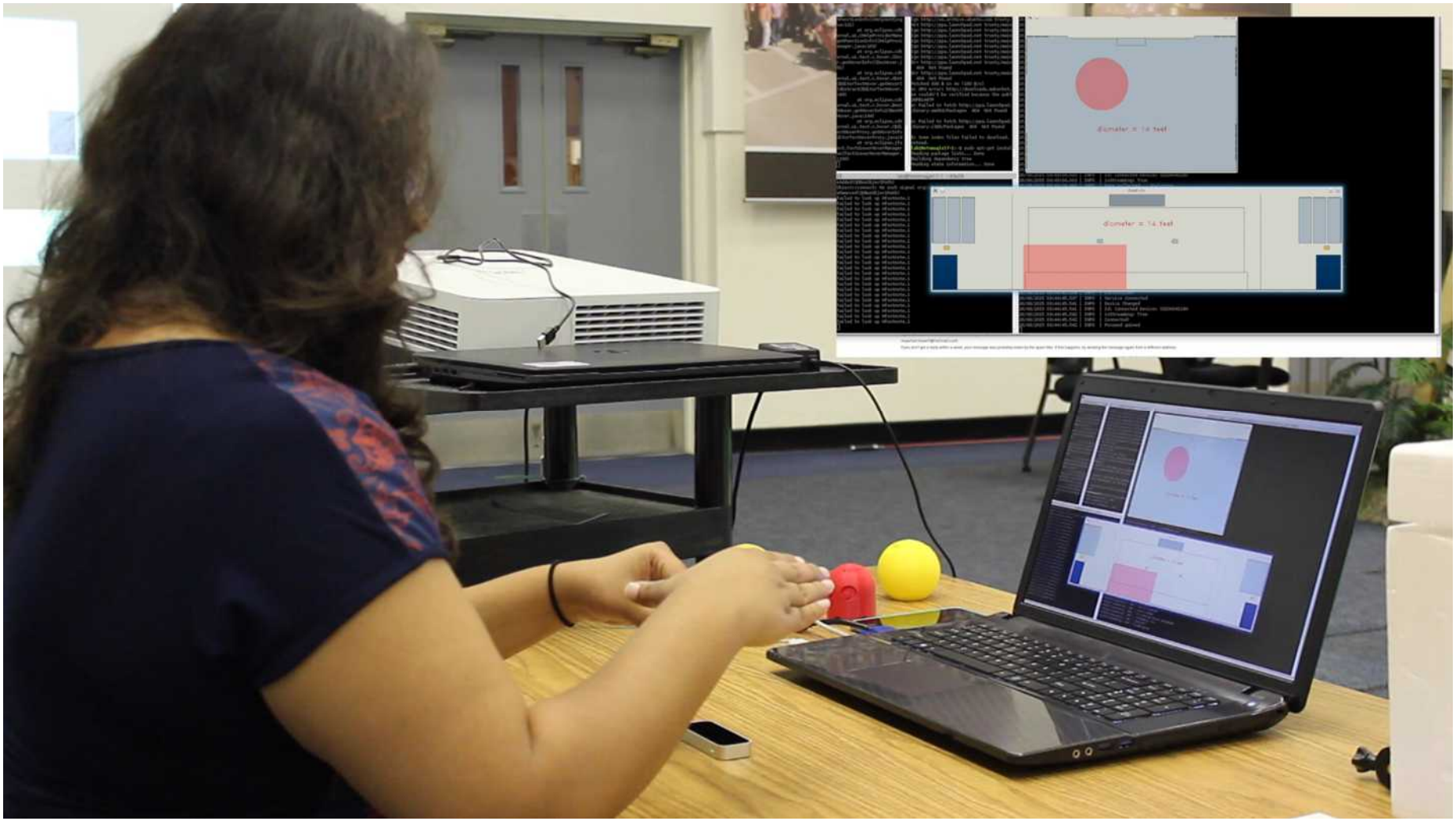


Autonomy Incubator



Technical Focus is…
safe reliable mobility and manipulation in dynamic, unstructured, and data-deprived environments.

Human Machine Teaming



Detect and Avoid

(#TreeDodging #DancesWithDrones)

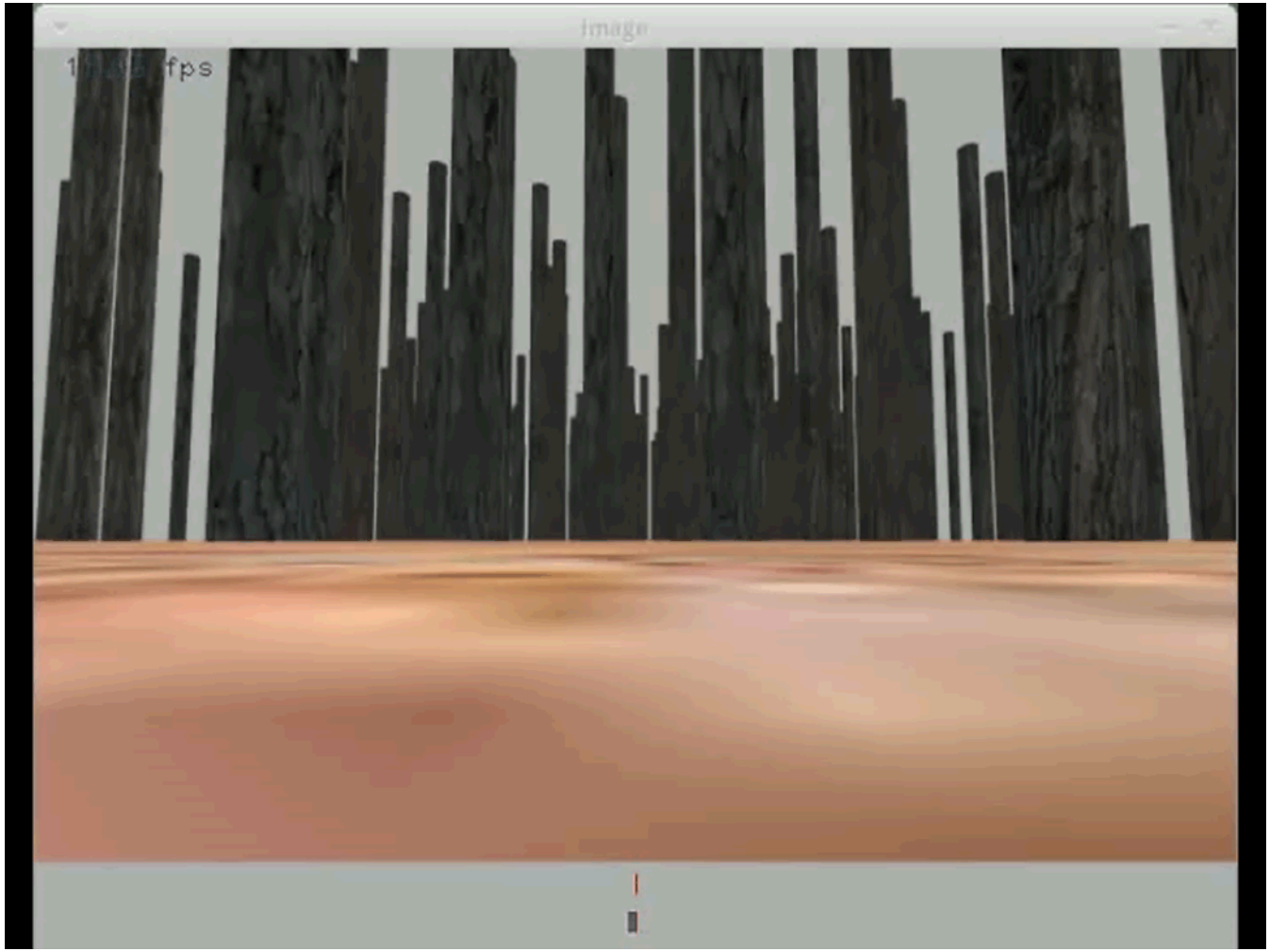


Detect and Avoid

(#TreeDodging)

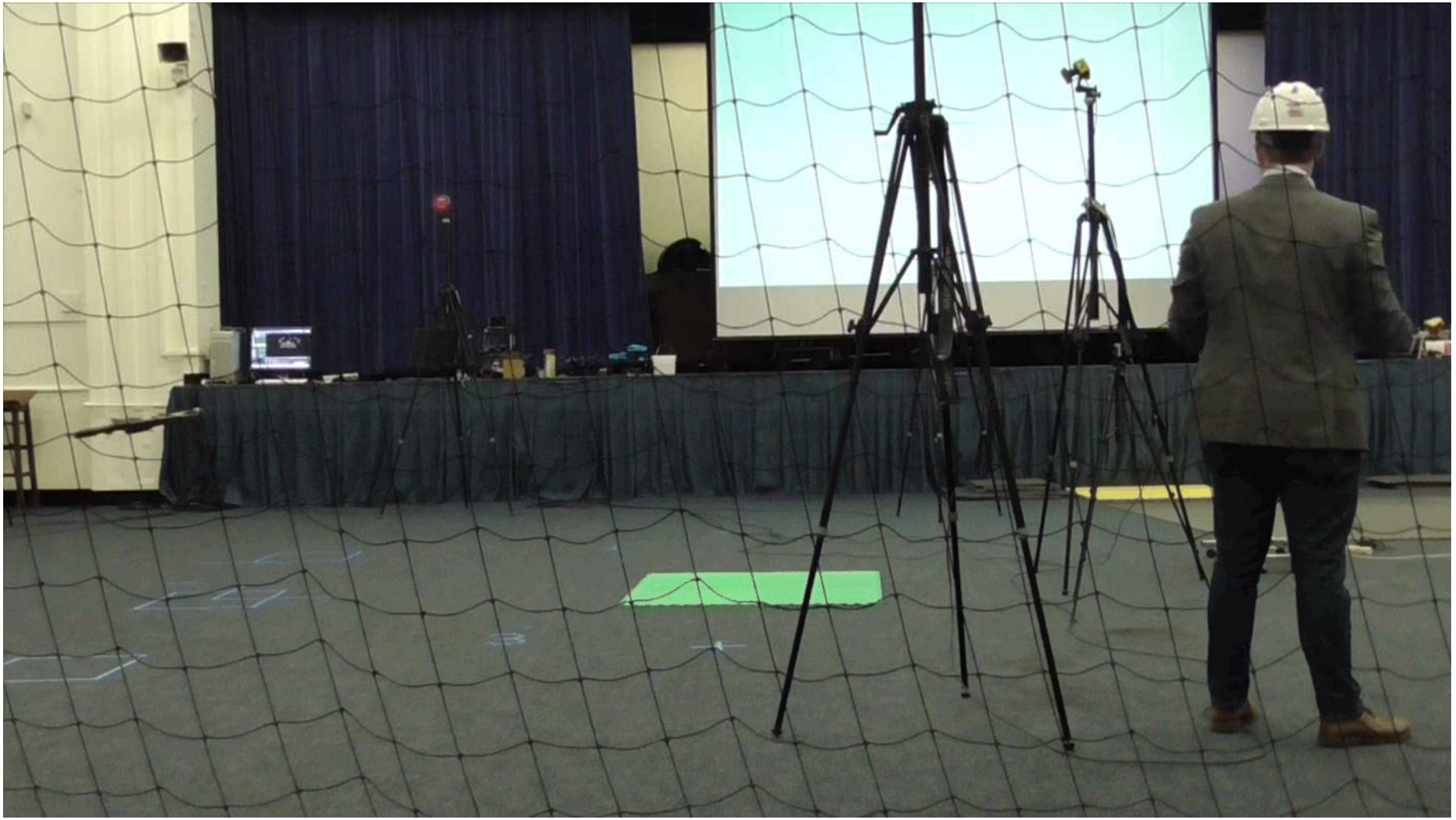


Machine Learning



Detect and Avoid

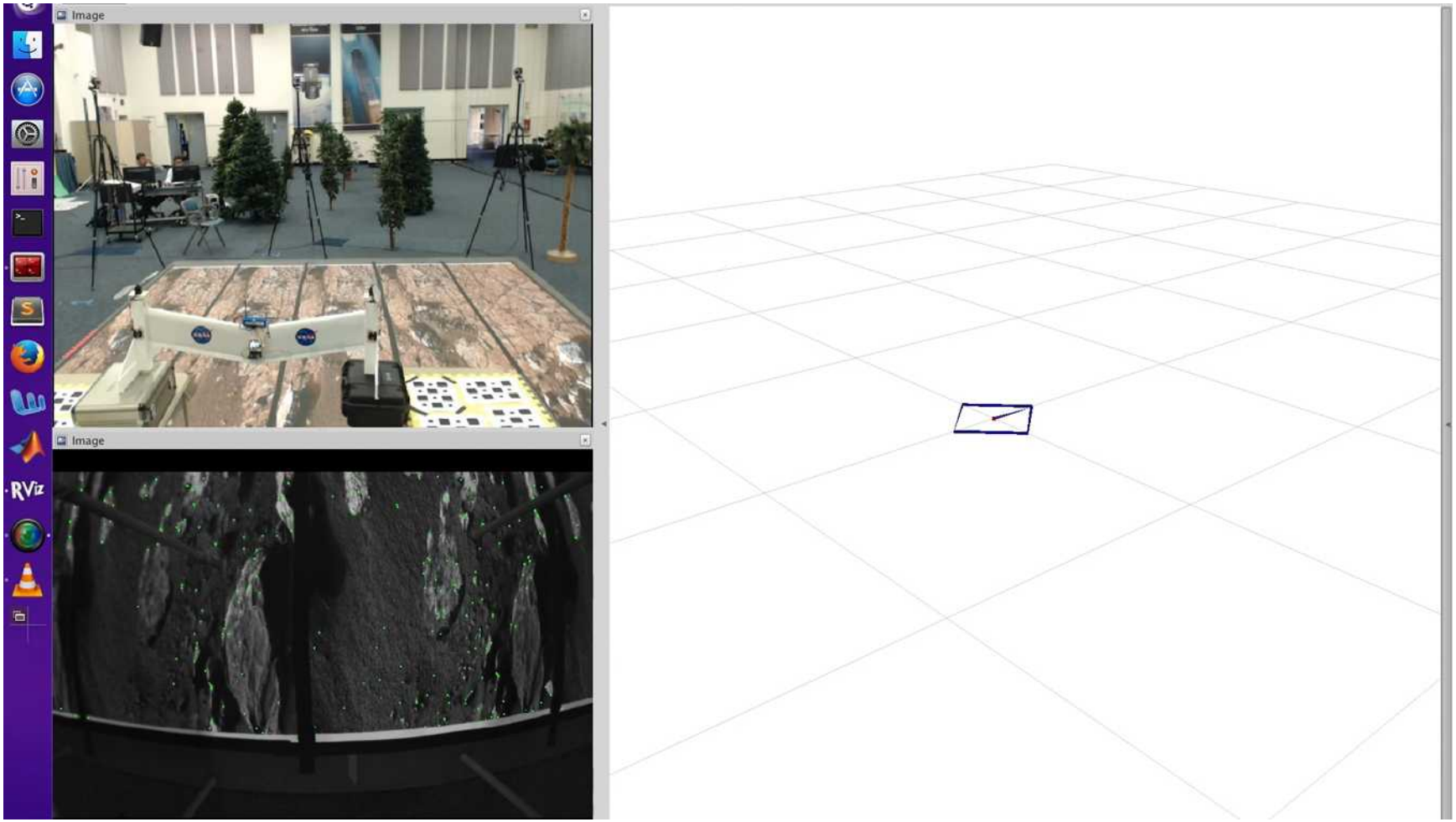
(#DancesWithDrones)



Precision Navigation



Robust Localization

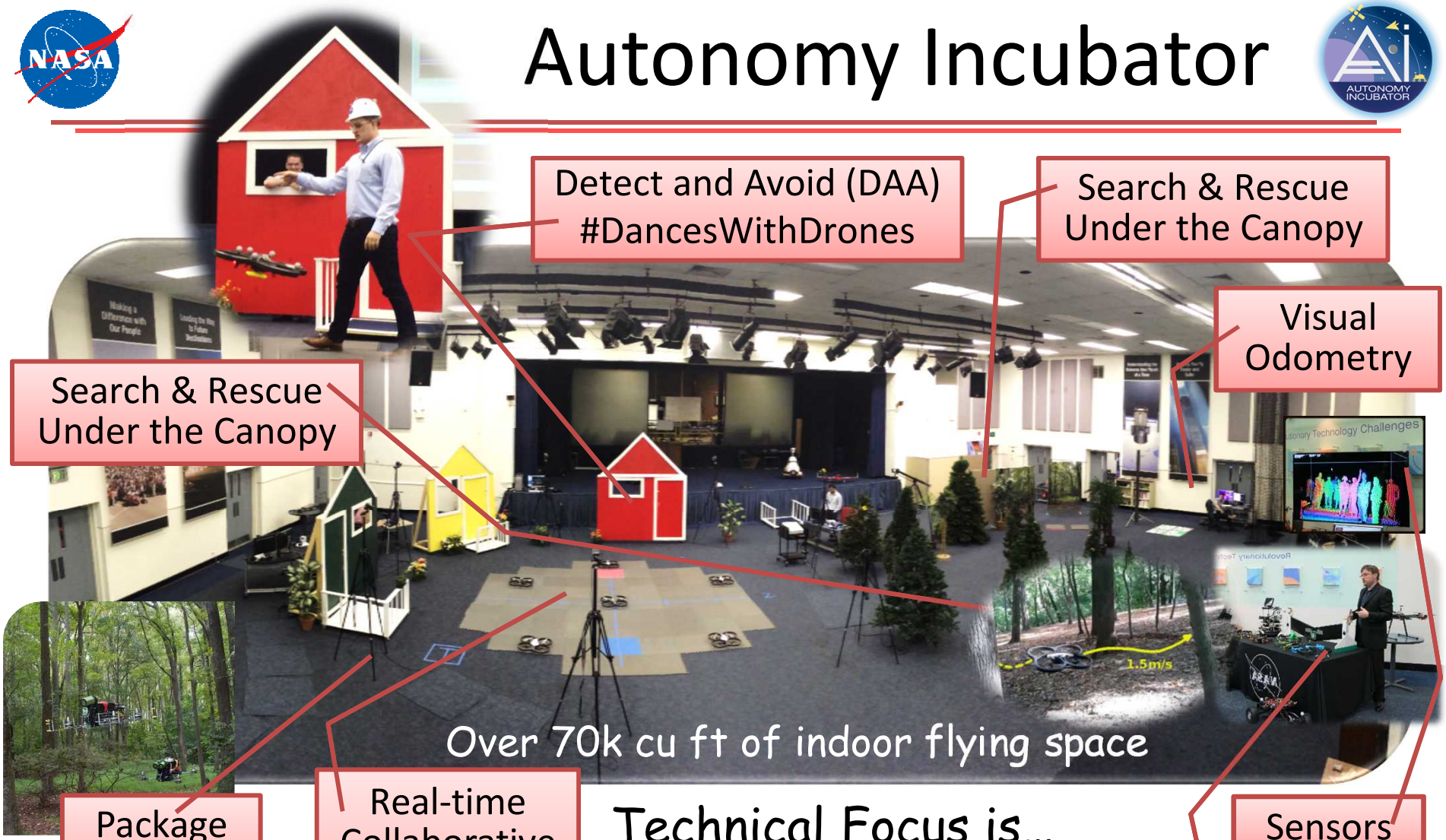


Test & Evaluation



CERTAIN - City Environment for Range Testing of Autonomous Integrated Navigation

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Earth Science Mission



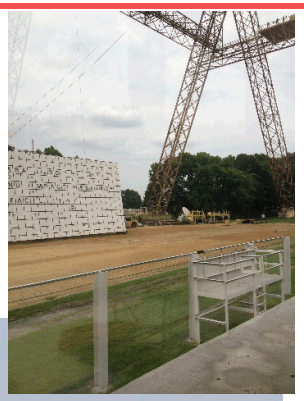
Air Quality Monitoring Sensors



- Multi-vehicle
- Data-denied

Search and Rescue

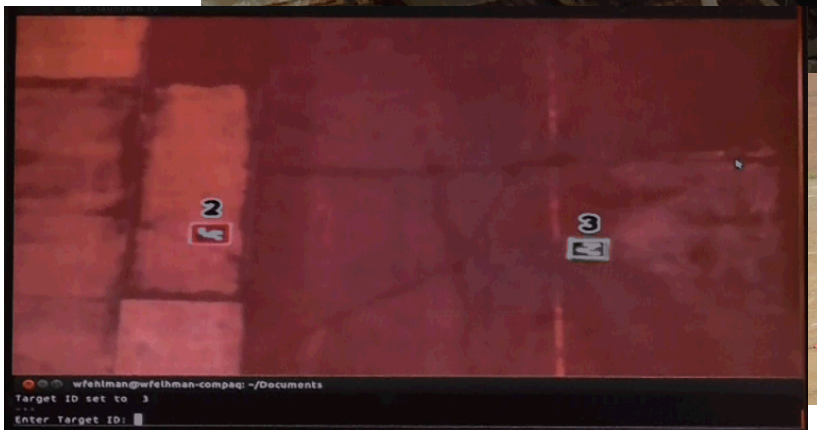














- Multi-vehicle
- Data-denied

Package Delivery



Thank You





B. Danette Allen
Senior Scientist for Autonomy
Head, Autonomy Incubator
danette.allen@nasa.gov